ASSIGNMENT 01

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BCA (AI & DS) - C

1. Explain the role of weights in a neuron.

Weights in a neuron determine the importance of each input feature. Every input is multiplied by a corresponding weight which controls its influence on output.

Larger weights indicate stronger influence, while small or negative weights reduce impact. During training, weights are adjusted to reduce prediction error.

In neural networks, weights act as learned parameters. Through backpropagation and optimization, the model continuously updates these values to improve accuracy.

Without weights, all inputs would contribute equally, making learning impossible. Therefore, weights help identify meaningful patterns from data.

2. What is an activation function?

An activation function is a mathematical function applied to the output of a neuron to determine final output.

It introduces non-linearity, enabling neural networks to solve complex real-world problems beyond simple linear relationships.

Common activation functions include Sigmoid, Tanh and ReLU. ReLU is widely used due to efficiency, while Sigmoid is common in binary classification.

Activation functions also help deep networks learn hierarchical patterns and improve decision making.

3. Define probability distribution in ML context.

A probability distribution describes how values of a random variable are spread and the likelihood of each outcome.

In machine learning, it helps represent uncertainty, model data behavior and support probabilistic predictions.

Examples include normal distribution, Bernoulli distribution and Gaussian distribution used in many ML models.

It is especially important in Naive Bayes, generative models and statistical learning.

4. What is gradient in optimization?

The gradient represents the direction and magnitude of the steepest increase of a function.

In optimization, algorithms like gradient descent move in the opposite direction of the gradient to minimize loss.

It helps update weights efficiently during training. Learning rate controls how large each update should be.

Gradient-based optimization is fundamental for improving neural network performance.